- Visualisation is shown in Figure 6.

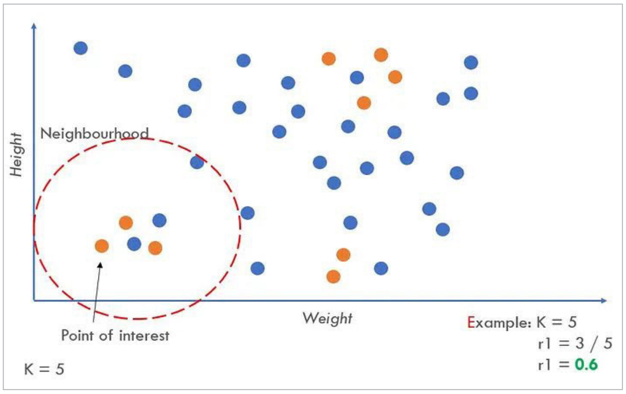

Figure 6: Adaptive Synthetic (Image Credit: *https://miro.medium.com/)*

- The sample code is:

```
from collections import Counter
from sklearn.datasets import make_classification
from imblearn.over_sampling import ADASYN
from matplotlib import pyplot
from numpy import where

X, y = make_classification(n_classes=2, class_sep=2,
weights=[0.1, 0.9], n_features=4, n_clusters_per_class=1,
n_samples=1000, random_state=10)

# summarise class distribution
print('Original dataset shape %s' % Counter(y))
Original dataset shape Counter({1: 894, 0: 106})

# perform upsampling
ada = ADASYN(random_state=42)
X_res, y_res = ada.fit_resample(X, y)
print('Resampled dataset shape %s' % Counter(y_res))
Resampled dataset shape Counter({0: 897, 1: 894})

# scatter plot of examples by class label
for label, _ in Counter(y_res).items():
row_ix = where(y_res == label)[0]
pyplot.scatter(X_res[row_ix, 0], X_res[row_ix, 1],
label=str(label))
pyplot.legend()
pyplot.show()
```

Pros and cons of upsampling

Pros

- Helps in increasing the under-represented class datasets.
- Reduces bias in the model.
- Cost-effective as it does not require additional dataset collection.

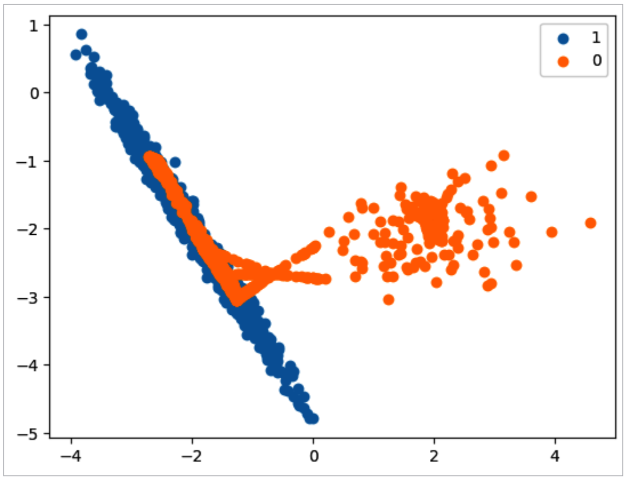

Figure 7: Scatter plot of examples by class label upsampled post using ADASYN

Cons

- Risk of overfitting when the synthetic data points closely resemble existing data points.
- The quality of synthetic data could negatively impact model accuracy.

We will conclude the article here to keep the learning concise. In the next article in this two-part series we will delve into downsampling. END

References

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- *https://imbalanced-learn.org/stable/references/under_sampling.html*
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